

NIKHIL SINGH

DevOps & Backend Engineer

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PROFESSIONAL SUMMARY

DevOps & Backend Engineer with **5+ years** designing and operating cloud-native platforms across **AWS and GCP**. Specialized in **Kubernetes (EKS + GKE)**, **GitOps (ArgoCD)**, **Terraform**, **Go microservices**, and **cloud cost optimization**. Built production systems serving **10M+ events/month** while cutting infrastructure costs, improving deployment velocity, and keeping revenue-critical services reliable.

CORE SKILLS

Cloud (AWS): EKS, ECS Fargate, EC2, Lambda, EventBridge, API Gateway, ALB, RDS (IAM auth), ElastiCache, MSK, S3, Route 53, Cognito, SQS/SNS, KMS, Secrets Manager, GuardDuty, IAM/IRSA

Cloud (GCP): GKE, Artifact Registry, Cloud SQL, Cloud Storage, BigQuery, Secret Manager, Cloud Build, Workload Identity

Kubernetes & Platform: multi-cloud EKS + GKE, Helm, ArgoCD (GitOps), Envoy Gateway (Gateway API, JWT), Kustomize, cert-manager, OPA

IaC & CI/CD: Terraform, AWS CDK, Serverless Framework, CloudFormation; GitHub Actions, Bitbucket Pipelines, Jenkins, OIDC federation, SHA-pinned images, Docker

Backend & Languages: Go (chi, gin, pgx, gRPC/protobuf), Python (FastAPI, boto3), Bash, PowerShell, SQL

Messaging & Streaming: NATS JetStream, Apache Kafka (Strimzi/MSK), RabbitMQ/AMQP, MQTT

Databases: PostgreSQL, MySQL, ClickHouse, Redis/Valkey, OpenSearch, DynamoDB; Flyway migrations, slow-query tuning

Observability & Reliability: OpenTelemetry, Prometheus, Grafana, Loki, CloudWatch, distributed tracing, incident response & root-cause analysis

Security & Payments: IAM least-privilege (IRSA), RDS/Valkey IAM auth, Cognito (OIDC/JWT/JWKS), KMS, Secrets Manager, WAFv2, ACM/TLS, multi-tenant isolation, Stripe billing

CERTIFICATIONS

AWS Certified Developer – Associate

IBM Docker Essentials

Oracle Cloud Infrastructure Foundations

IBM SQL

PROFESSIONAL EXPERIENCE

Good AI Labs — DevOps & Backend Engineer

Apr 2026 – Present

- Own **multi-cloud CI/CD and GitOps delivery** for a real-time sports-odds and order-book trading platform of **~19 Go microservices on AWS EKS and GCP GKE**, where every merge ships an immutable SHA-pinned image through **ArgoCD** behind a GitHub Environments approval gate.
- Built **runtime-configurable bet-placement rate limiting** for the order-book service — operator-tunable caps persisted in Postgres and mirrored to Redis, returning standards-compliant Retry-After guidance via gRPC RetryInfo — validated under a **10,000-wager burst test**.
- Engineered a **durable statistics and timeline projector** over the NATS livescore stream using deterministic content-hash incident IDs and safe in-place deduplication migrations, reducing a live match page from **667 noisy rows to 54 event-grade rows**.
- Migrated **9 Go services** from static database passwords to **short-lived AWS RDS IAM tokens** minted per connection through a pgxpool hook with TLS enforced end to end, consolidating three roles across four databases into a single least-privilege **IRSA-bound IAM role**.
- Built the **Stripe billing backend** for Verso, an AI-assisted coding-assessment SaaS — plan upgrades with immediate invoicing and receipts, payment-failure detection with customer notification emails, and safeguards against incomplete-subscription pile-up and stale downgrades.
- Hardened Verso's **proctored candidate sandbox** (VS Code on ECS Fargate and GitHub Codespaces) with extension-install lockdown, disabled AI assistants, an escalating countdown timer, and a clean early-finish flow that seals sessions as complete.
- Replaced a hand-rolled schema self-healer with **versioned SQL migrations**, and added a **post-deploy health smoke test** to CI so a deployment that cannot boot fails its own pipeline run.
- Built an automated **cloud-cost reporting service** on AWS Lambda and EventBridge that renders per-service PDF cost reports with color-coded deltas into Slack, surfacing a **3x AWS bill increase** from cross-AZ transfer and a GCP logging sink quietly costing **~\$2K/month**.

Enetro (Readywire Subsidiary) — Cloud Automation & AI Platform Engineer

Jan 2026 – Jul 2026

- Architected a reusable, **module-driven AWS platform in Terraform** covering VPC, ALB, ECS Fargate, API Gateway, RDS, and Route 53, with isolated per-environment state that eliminated infrastructure drift across **four environments** and underpins **40+ event-driven Python Lambda microservices**.
- Engineered **zero-credential CI/CD** on Bitbucket Pipelines using **OIDC federation**, delivering secure ECR-to-ECS rolling deployments with Secrets Manager injection that take a code push to a hands-off production rollout in about **5 minutes**.
- Designed a multi-service **ECS Fargate architecture** separating a request-rate-scaled **FastAPI tier** behind an ALB from an SQS-queue-depth-scaled **Celery worker tier**, enabling independent scaling and zero-downtime rolling deployments.
- Built an in-house **observability platform** on AWS serverless primitives — a Lambda OTLP proxy and an OpenTelemetry Collector on Fargate — processing **10M+ events per month for under \$80** with end-to-end distributed tracing, replacing Datadog and New Relic.
- Implemented **multi-tenant isolation on AWS Cognito** with Terraform-managed user pools, JWT enrichment through a Pre-Token-Generation Lambda, and a custom API Gateway authorizer validating signatures against JWKS, on top of a least-privilege IAM and customer-managed KMS baseline.

- Operated a **production AI pipeline** for transcription, translation, and LLM feature extraction, using LiteLLM for provider-agnostic routing, Instructor for schema-validated output, and Langfuse for per-tenant token-cost attribution.

Readywire — DevOps Engineer

Jun 2023 – Jan 2026

- **Improved database performance by 80%** with an event-driven RDS optimization pipeline in which AWS Lambda toggles slow-query logging on CloudWatch CPU-spike alarms without a database restart, enabling immediate performance diagnostics.
- **Accelerated application delivery by 75%**, from 3 hours to 45 minutes, by architecting an end-to-end Jenkins CI/CD pipeline for Acumatica using PowerShell and REST APIs.
- **Reduced manual database operations by 65%** by orchestrating secure MySQL backup pipelines in Bash featuring encrypted credential handling, on-the-fly 7z compression, schema validation with retries, and hands-off delivery to S3 with pre-signed download URLs.
- Created an **operational tooling library** in Python and Bash, including a cross-account S3 migrator with dry-run support, a 16-thread parallel mydumper/myloader RDS shuttle, and a weekly GuardDuty findings reporter for the security team.
- Hardened production resilience by automating **RDS snapshot backups with retention and scheduled restore drills**, alongside unattended EC2 patching and volume lifecycle management.

Tata Consultancy Services — Associate Engineer

Aug 2021 – Jun 2023

- Provided **L1/L2 production support** on a rotational on-call schedule, resolving incidents within **SLA** and escalating critical issues through **ServiceNow** with clear root-cause documentation.
- **Automated user onboarding** and account provisioning in Python, replacing static Excel trackers with a self-service workflow and cutting new-hire setup time **from hours to minutes**.

PROJECTS

Cloud Resume

codenickk.com

- Serverless resume website hosted on **S3 and CloudFront**, with a **Lambda + API Gateway + DynamoDB** visitor counter.
- **GitHub Actions CI/CD** deploys every push and invalidates the CDN cache automatically.

FuelMeter (PWA)

fuelmeter.codenickk.com

- Offline-first fuel-log PWA built with **React, TypeScript, and Vite** that computes brim-to-brim km/L, cost per km, and monthly spend.
- localStorage-first data layer with **Supabase** sync behind an auth gate, served from S3/CloudFront.

Shramik

github.com/0NikhilSingh5/shramik-io

- **Android app** and public web form for registering informal workers in India, backed by **Supabase** in ap-south-1 for PII residency.
- Signed releases with an in-app updater; the entire system runs at **₹0/month**.

EDUCATION

B.Sc. Computer Science — Invertis University, Bareilly

2018 – 2021